Machine Learning for Waste Management

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[Our Project!]

Research Problem: Classifying Waste with Deep Learning

How effectively can modern deep learning models classify different types of waste—and do these models generalize well enough to support real-world automated recycling systems?

1. Identifying E-Waste: A Critical Challenge

Electronic waste (e-waste) is a growing global concern. Effective classification is the first step towards efficient recycling and resource recovery. Our project focuses on developing machine learning models to accurately identify various e-waste components, paving the way for automated sorting systems.

The Data Behind E-Waste Classification

Dataset Overview

Dataset via Kaggle: [E Waste Image Dataset]

Last Updated : 2023

10 distinct classes: including common items like Battery, Keyboard, PCB, and Mobile Phone.

Original Splits: The dataset came with predefined **train, test, and validation sets.**

My Usage: We merged the original test and validation sets into a single test set for a more balanced evaluation. The images are typically older, with pretty plain light to medium toned backgrounds and minimal variation within each class

Hand made Validation for Real-World Scenarios

To better assess real-world performance, we created a custom validation set:

- **10 images per class:** A total of 100 images.
- **Manual curation:** Images were carefully selected to represent **diverse class types, older and newer photos**, lighting, angles, and rougher less intuitive conditions not typically found in the original dataset.

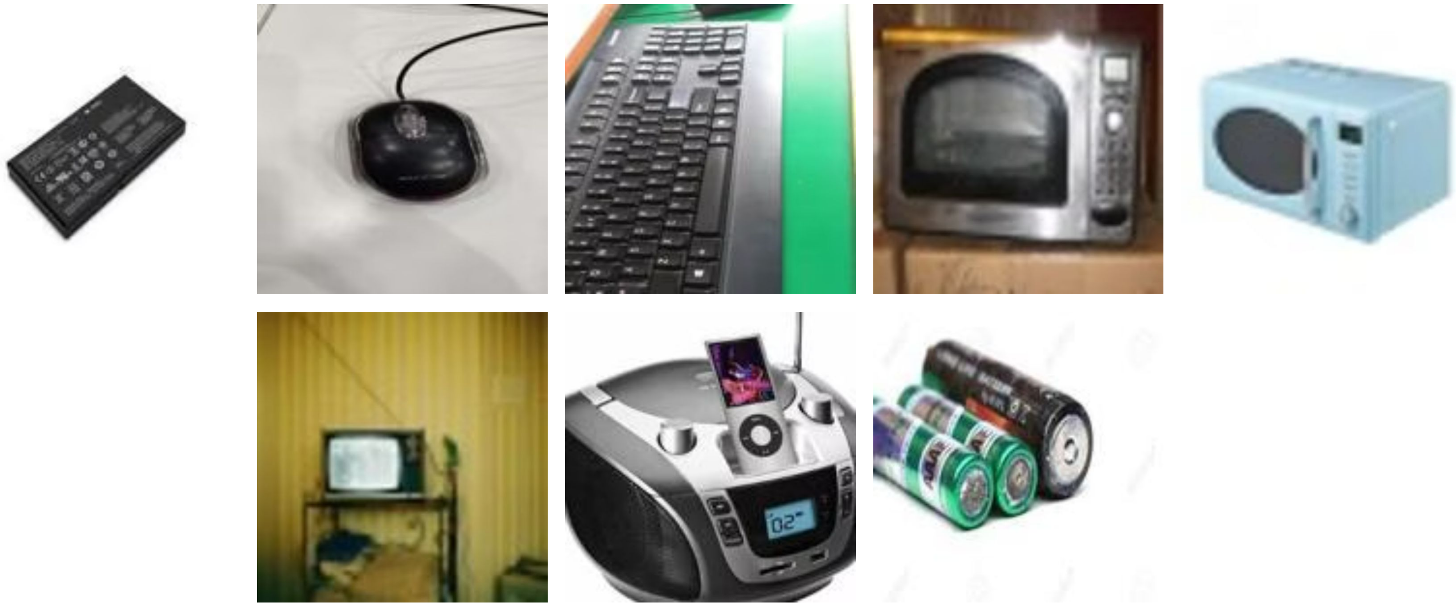


Image Differences

Kaggle Data

Hand made Validation Data

Class : Battery

Model Construction

1. Custom Convolutional Neural Network (CNN) V3

- **3 convolutional blocks** with increasing filters
  (16 → 32 → 64 → 128)
- **BatchNorm** in early layers for stabilizing training
- **ReLU** activation throughout
- **MaxPooling** after major blocks to reduce spatial size
- **Classifier:** Flatten → Linear(256) → ReLU → Dropout(0.15) → Linear(10)
- Trained with **Adam** (lr = 8e-4) and **CrossEntropyLoss**
- **Batch size: 32 | Epochs: 30**

2. Transfer Learning: ResNet18

- **Pretrained on ImageNet** to leverage strong general feature extraction
- All base **layers frozen** to preserve pretrained weights
- **Final fully connected layer** replaced to output **10 e-waste classes**
- Benefits: Only the new classifier was trained → **faster and more stable training**
- Trained with **Adam** (lr = 1e-3) and **CrossEntropyLoss**
- **Batch size: 32 | Epochs: 20**

Comparative Results: Custom CNN vs. ResNet18 on E-Waste

1. Custom CNN (V3) Performance

Final Test Accuracy: 67%

Best Training Accuracy: 96.58%

Best Test Accuracy: 67.00%

Main Observation: This model showed significant **overfitting.** It frequently confused visually similar classes and struggled with inconsistent image quality and lighting, indicating slower learning without pre-trained features.

2. Transfer Learning: ResNet18 Performance

Final Test Accuracy: 91.8%

Best Training Accuracy: 92.46%

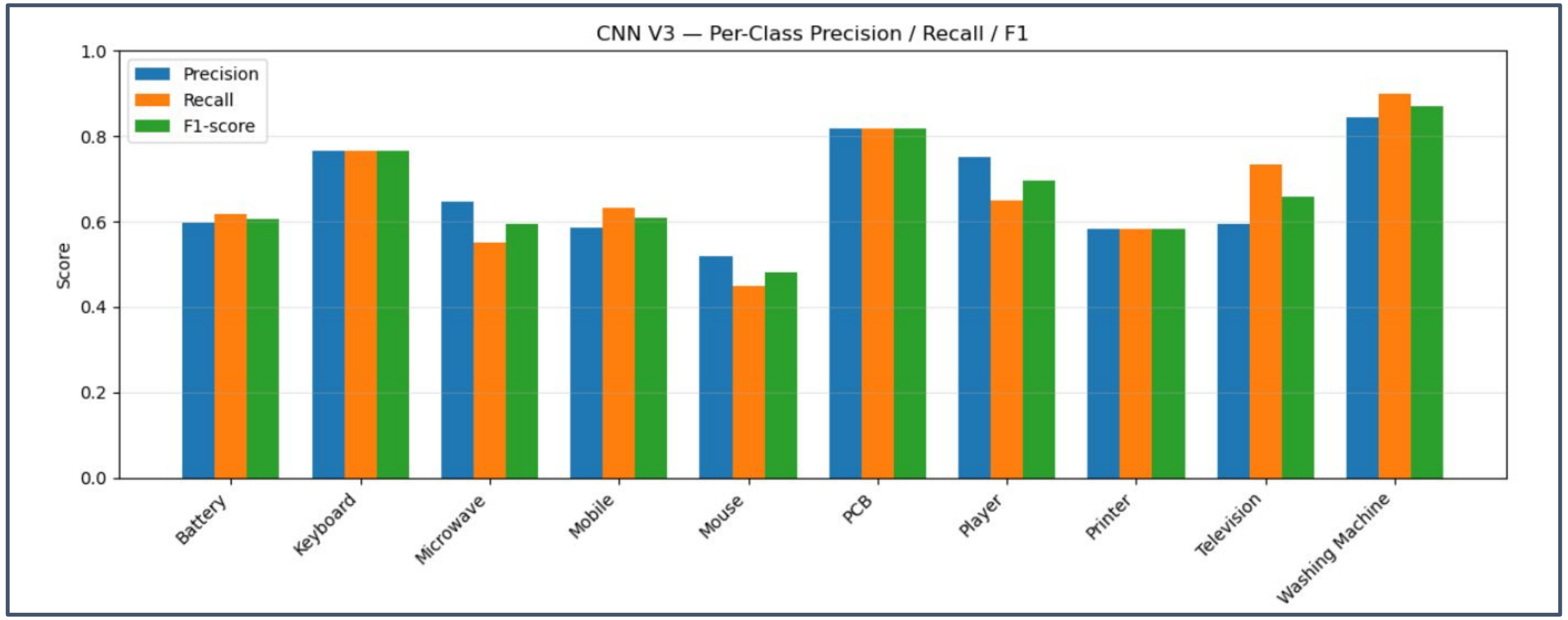
Best Test Accuracy: 92.00%

Main Observation: ResNet18 converged quickly due to its pre-trained features. It exhibited **minimal overfitting**. This model performed significantly better on challenging classes and ambiguous items (e.g., Printer vs. Player, Television vs. Microwave) and showed greater robustness to similar shaped objects and older image conditions.

Model Performance: Custom CNN vs. ResNet18 on E-Waste

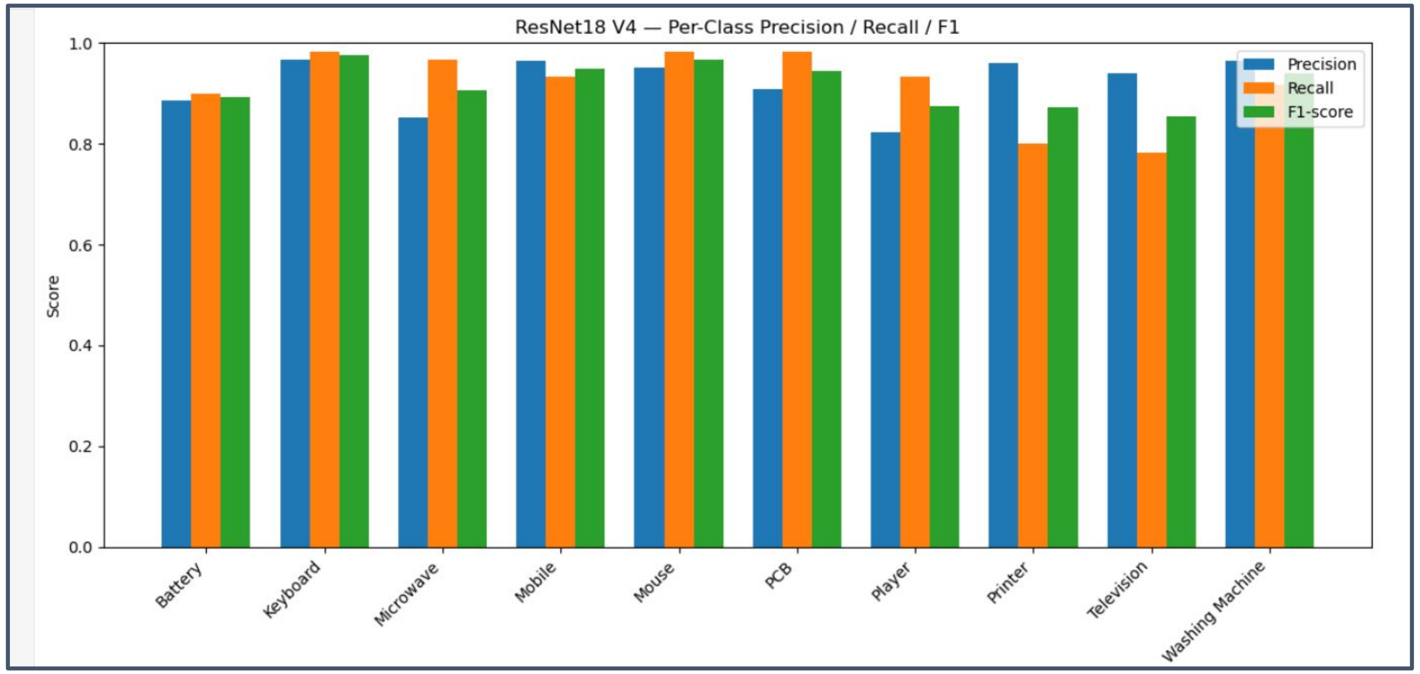
1. Custom CNN (V3) Performance

- **Strongest classes:** PCB, Keyboard, Washing Machine
- **Weakest classes:** Mouse, Microwave, Television
- Shows uneven performance and struggles with class similarity + noise



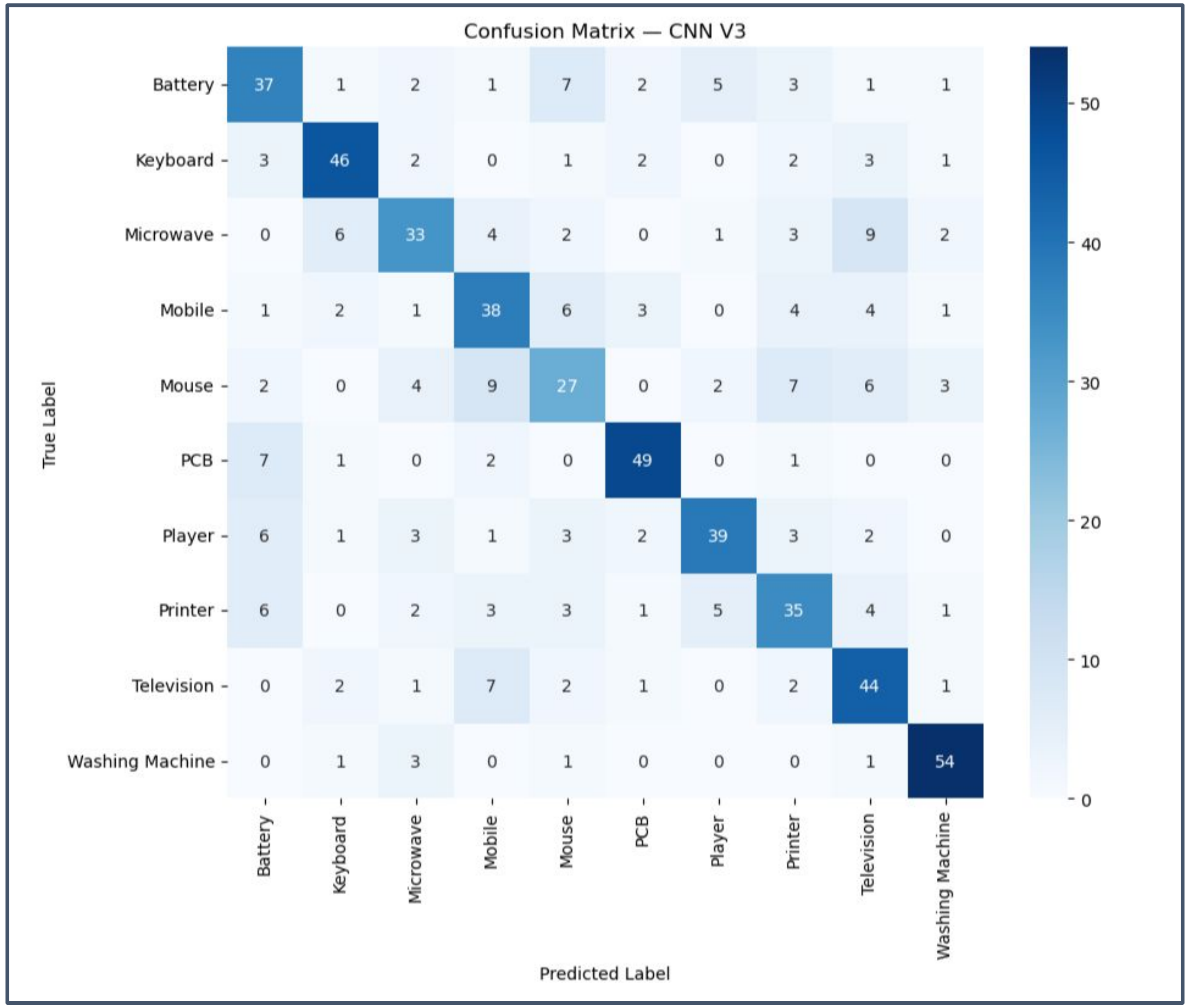
2. Transfer Learning: ResNet18 Performance

- High accuracy across **all 10 classes**, especially small/difficult items
- Best overall performance on **Keyboard, Mouse, Mobile, Washing Machine**
- Shows **strong generalization** due to pretrained ImageNet features

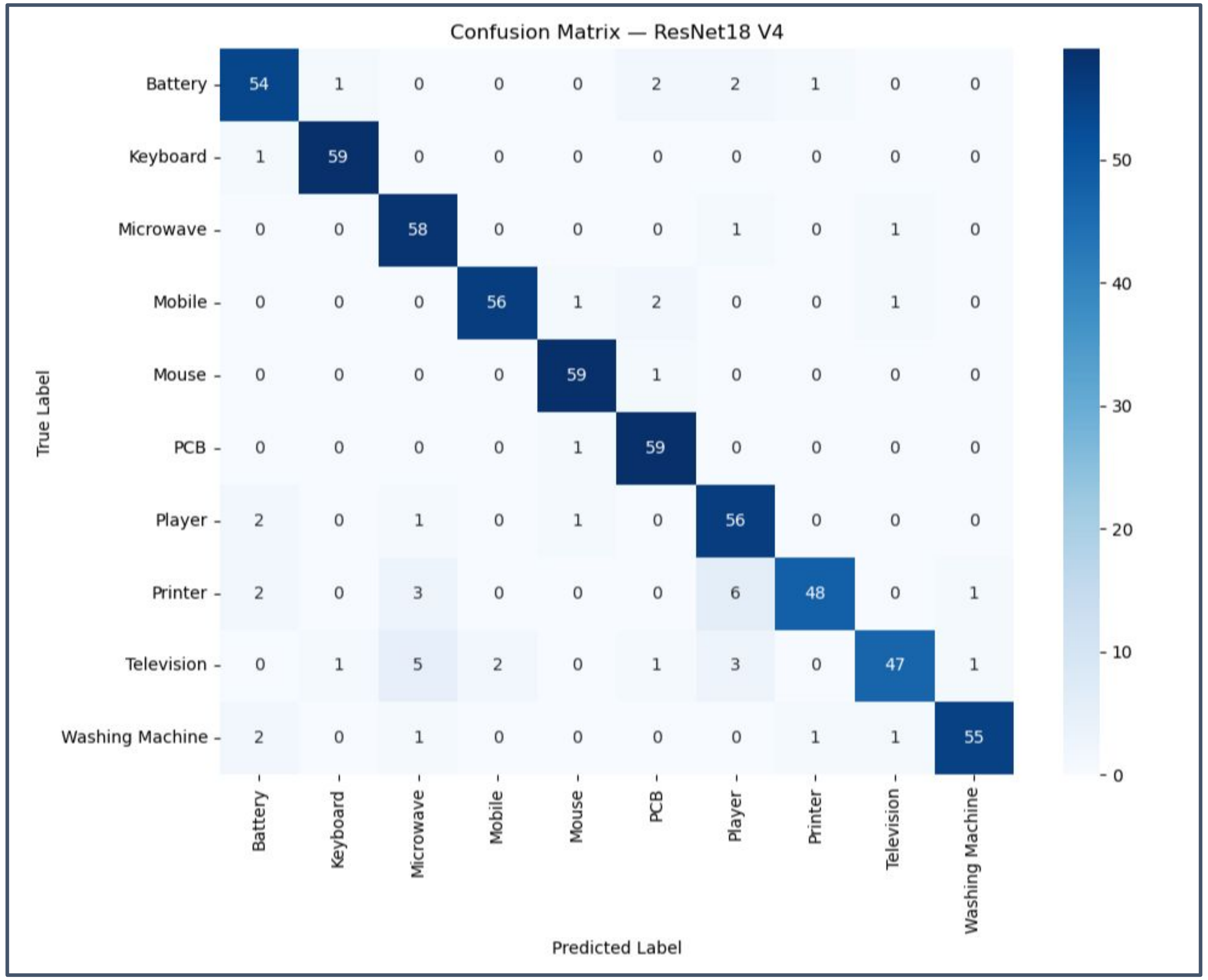


Model Performance: Custom CNN vs. ResNet18 on E-Waste

1. Custom CNN (V3) Performance



2. Transfer Learning: ResNet18 Performance



Model Performance: Custom CNN

Model Performance: ResNet18

Bonus: Real-World Validation and Key Takeaways

Custom External Validation Set

Composition: 100 new images, with 10 per e-waste class.

Sourcing: Images were gathered online to simulate diverse real-world scenarios.

Purpose: To introduce purposeful variation in electronic variations within each category, lighting, and angles, providing a more rigorous test of generalization.

Model Performance on External Data

Custom CNN (V3): 30.6% accuracy

ResNet18: 60.2% accuracy

ResNet18 demonstrated superior stability and generalization, even with unseen real-world variations.

Key Takeaways

Accuracy Drop

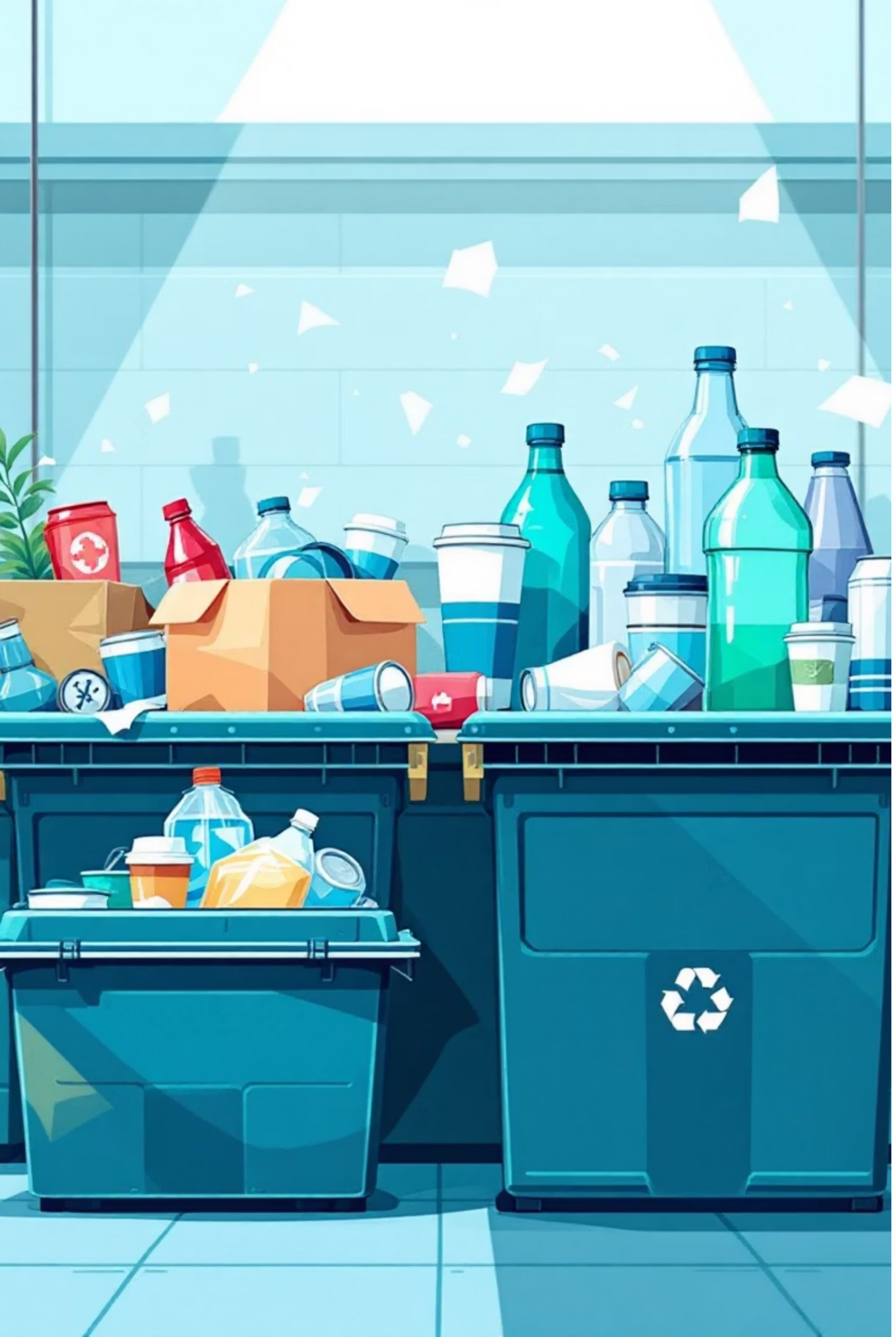
Both models experienced a notable drop in accuracy on the custom real-world images, highlighting the challenge of generalizing to truly novel data.

Custom CNN's Struggles

The custom CNN struggled significantly with class similarity and environmental noise, revealing its limitations without extensive pre-training.

ResNet18's Robustness

ResNet18, leveraging pre-learned features, handled background variations and diverse conditions far better, indicating its potential for practical applications.



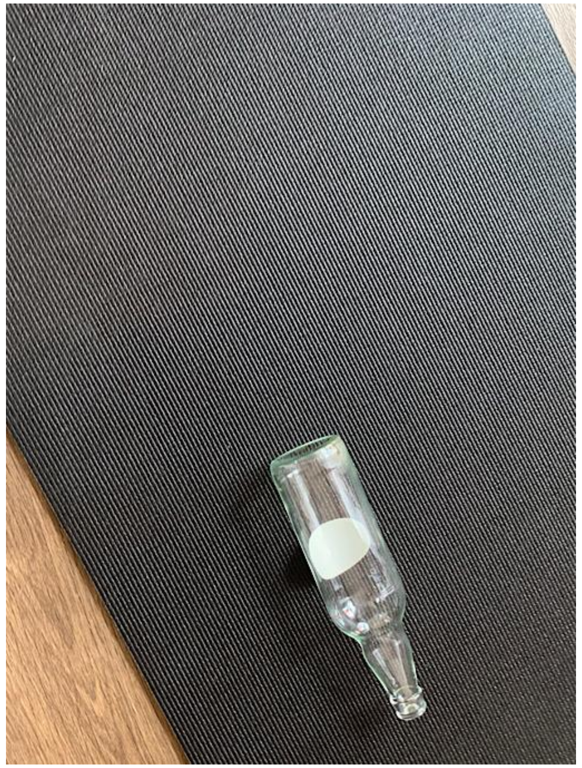
Detecting Drinking Waste: A New Challenge

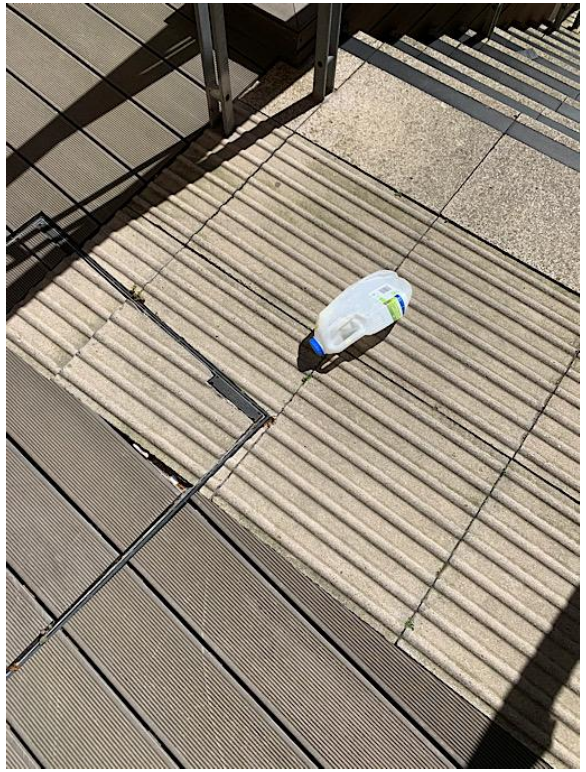
Accurate detection of recyclable drinking containers is essential for automated sorting and recycling processes in public spaces and commercial facilities.

Drinking Waste Detection Data

Kaggle Drinking Waste Classification Dataset

- **4 Classes** including Aluminum cans, plastic bottles, glass bottles and milk jugs
- Over 4000 images with classes equally represented
- Contains YOLO style bounding boxes
- Split with random split for training and testing
- Fairly simple images

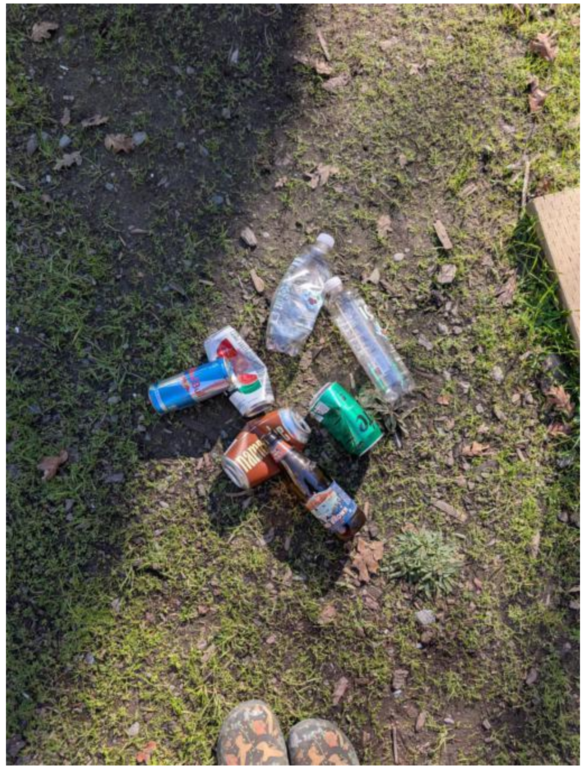




Drinking Waste Detection Data

Handmade Validation Set

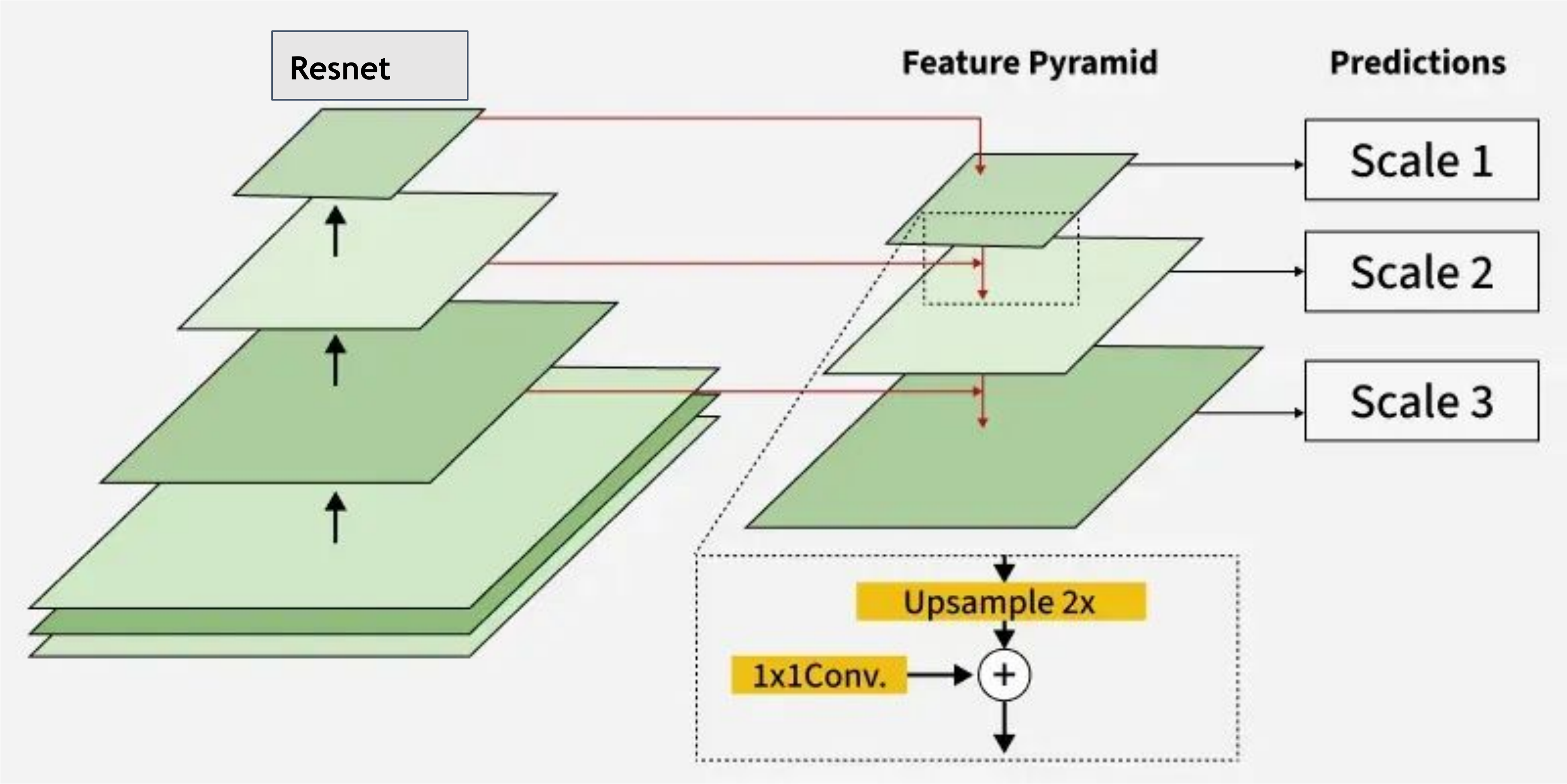
- ‘Messier’ images to reflect real world scenarios
- Includes busy backgrounds, overlapping objects, partially obscured objects, and objects that aren’t targets
- Combination of personal photos and internet sourced photos
- Annotated with MakeSense
- Renamed and resized for consistency



Model Architecture

Custom Backbone FasterRCNN

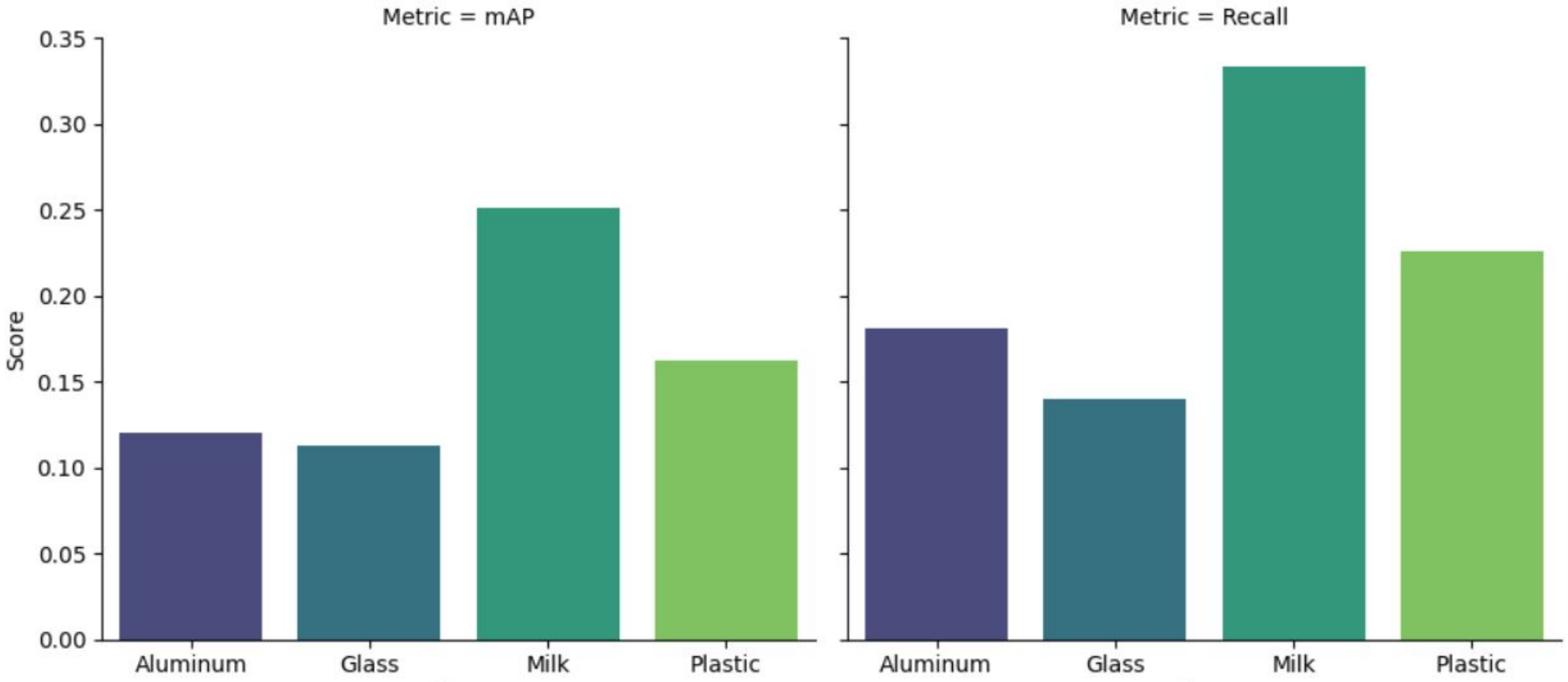
- Backbone has 5 resnet layers feeding 3 feature pyramid layers
- Combines the power of resnet with the multi-scale prediction ability of feature pyramids
- Uses SGD optimizer
- Requires data be a custom dataset class



Model Performance

mAP and Recall Scores

- Extremely low mAP (< 0.3) and recall (< 0.35)
- Much better mAP scores at lower IOU thresholds
- Indicates failure at higher IOU thresholds and overfitting to training data

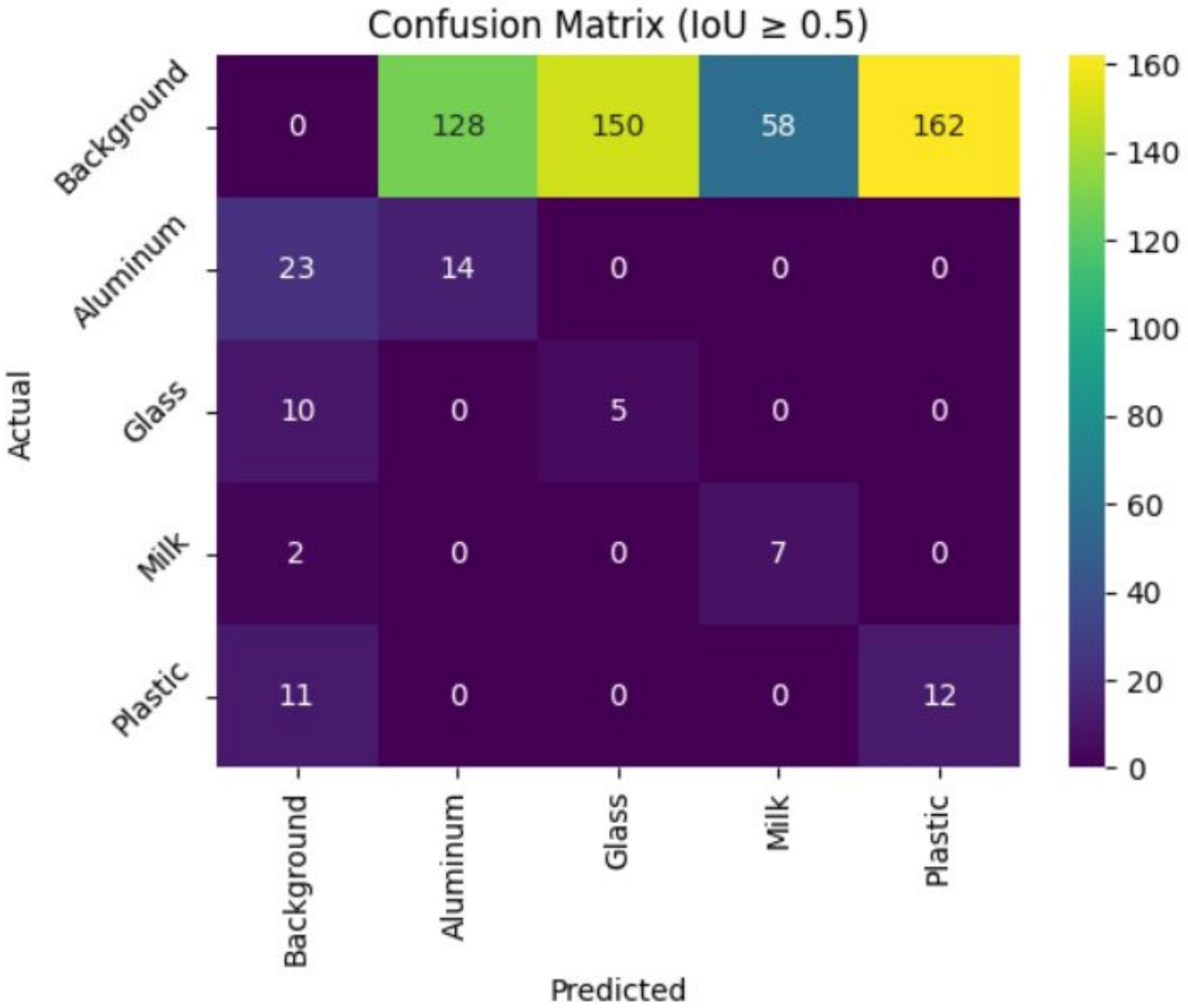


Model Performance

Confusion Matrix

- Similar matrix results on both sets
- Considers the highest score prediction within the IOU threshold of a target
- Good at not misidentifying one target as another
- Bad at falsely identifying objects where there is only background

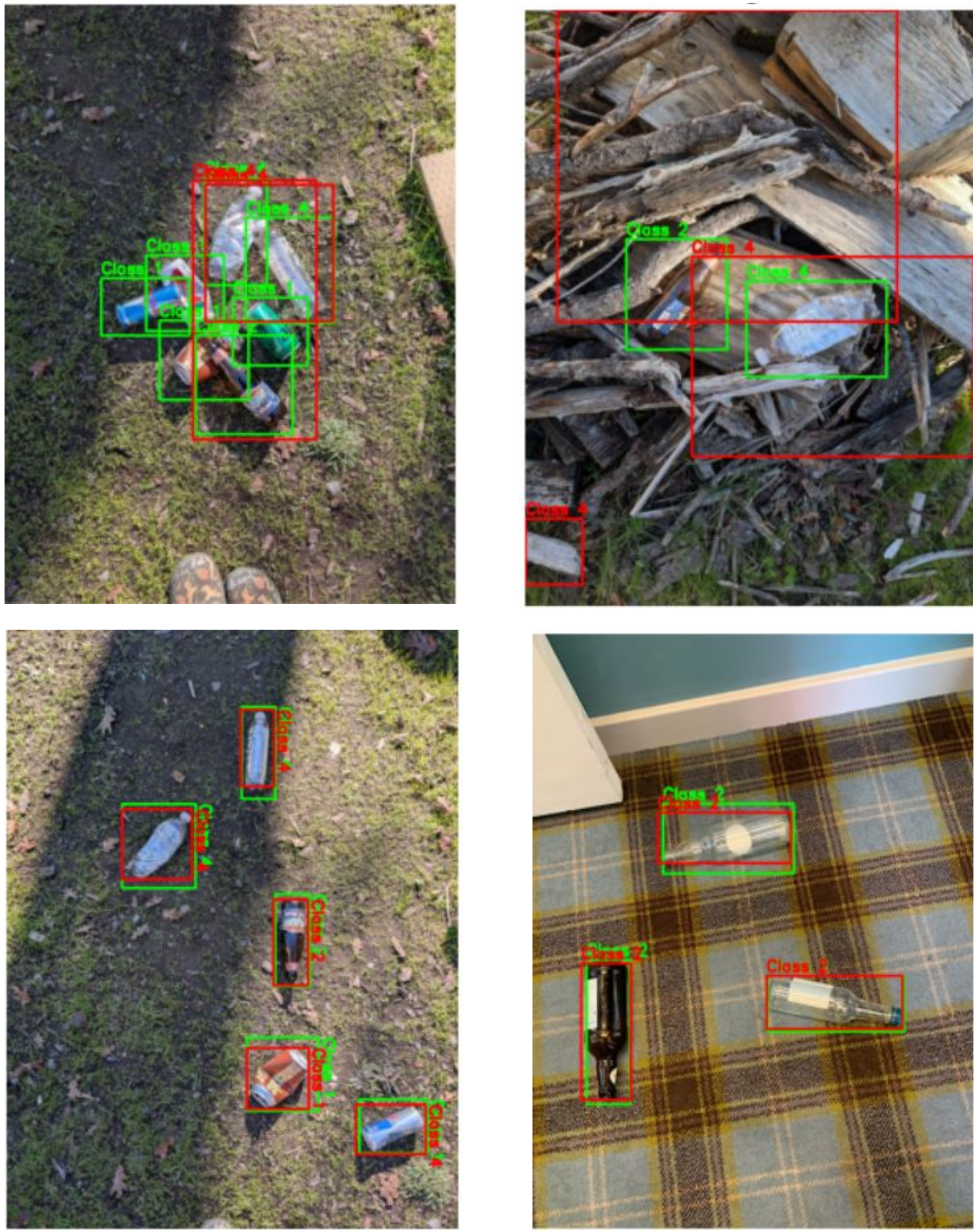
Validation



Model Performance

Predicted Boxes

- Excels at detecting well spaced objects on simple backgrounds
- Fails to detect clustered objects
- Fails to detect objects on busy backgrounds
- Predicted boxes are filtered by IOU threshold = 0.5 and probability threshold = 0.5



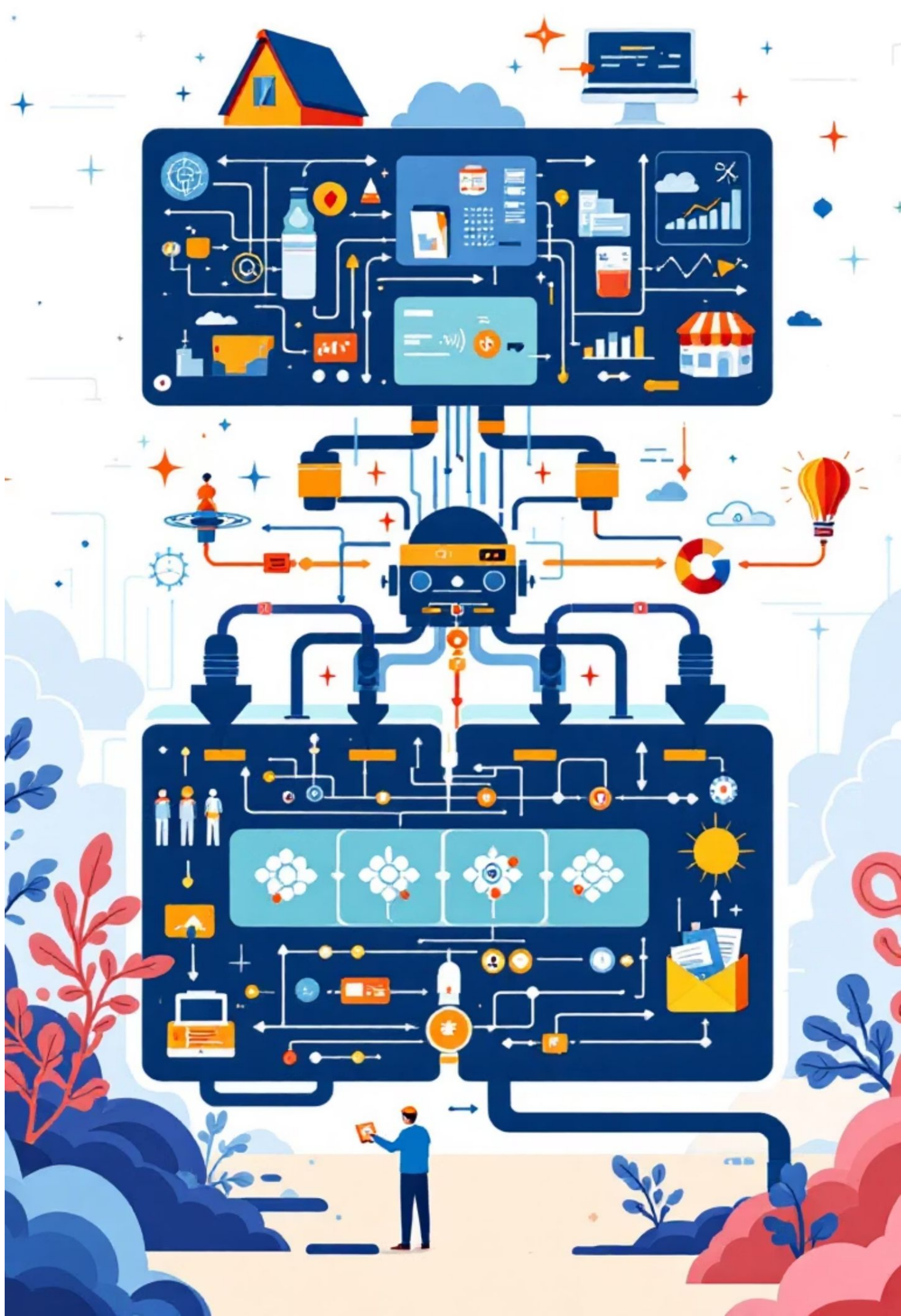
Model Performance

Takeaways

- Great performance on highly controlled images
- Unable to effectively generalize to real world images
- Next step is to improve model with more robust training data
- Model is not ready for real world deployment but initial findings show that there is strong potential for an improved model to effectively sort drinking waste

Conclusion

- Models can classify and detect waste effectively, but **performance depends heavily on data quality**.
- Model accuracy is dependent on **robust training data**
- Models trained on uniform data struggle with **real world variation**
- Misidentified objects are often **visually similar** to other classes or have **complicated backgrounds**
- **ResNet** is a powerful for achieving better generalization
- ML is promising for automated waste sorting, but **reliable deployment requires diverse data and thorough evaluation**



Thank You!